

TABLE OF CONTENTS

ACKNOWLEDGEMENT

CHAPTER 1	5
INTRODUCTION	5
1.1 Project Background/Problem Statement/ Business Problem	5
1.2 Objectives of the project	5
1.3 Overview/Scope of the project	Error! Bookmark not defined.
1.4 Overview of the Methodology.....	3
1.5 Benefit of the project	Error! Bookmark not defined.
CHAPTER 2	8
DATA PREPARATION AND PREPROCESSING	Error! Bookmark not defined.
2.1 Dataset and Data Statistics	Error! Bookmark not defined.
2.2 Data Preprocessing	Error! Bookmark not defined.
2.2.1 Handling Missing Values.....	5
2.2.2 Data Transformation.....	5
2.2.3 Feature Selection.....	5
2.2.4 Data Normalization.....	5
2.3 Data Visualization.....	5
CHAPTER 3	Error! Bookmark not defined.
APPLIED METHODOLOGIES AND MINING	Error! Bookmark not defined.
3.1 Descriptive Mining	Error! Bookmark not defined.
3.2 Modeling	Error! Bookmark not defined.
3.2.1 Applied Methodologies /Algorithms.....	5
3.2.2 Model Set up.....	5
3.2.3 Baseline Model.....	5
3.2.4	5
CHAPTER 4	Error! Bookmark not defined.
EVALUATION	Error! Bookmark not defined.
4.1 Performance Metrics	Error! Bookmark not defined.
4.2 Comparison of Experiment Results/Model Comparison	Error! Bookmark not defined.
4.2.1 Evaluation Analysis.....	6

4.2.2 ROC Curve and AUC.....	6
4.2.3 Cross Validation	6
4.2.4 Findings.....	6
CHAPTER 5.....	Error! Bookmark not defined.
CONCLUSION	Error! Bookmark not defined.
5.1 Advantages of using this project	Error! Bookmark not defined.
5.2 Disadvantages of using this project/ Limitation	Error! Bookmark not defined.
REFERENCE	

LIST OF FIGURES

Figure 1. 1 Caption16

Figure 3. 1 Caption..... **Error! Bookmark not defined.**

LISTS OF TABLES

Table 4. 1 Table Caption..... **Error! Bookmark not defined.**

ACKNOWLEDGEMENT

At the outset, I wish to thank my advisor, Professor **Dr. Daw Hus Myat Mo**, for the useful guidance, professional inputs, and continuous encouragement in developing this project. My understanding of Data and Knowledge Mining, and the continuous feedback and encouragement from her have been helpful in guiding this research effort on deforestation prediction and analysis.

This project has used data related to deforestation and therefore, I wish to thank for the availability of comprehensive environmental and socio-economic data that facilitated the study.

Lastly, I wish to thank the **University of Computer Studies, Yangon (UCSY)** for offering the appropriate academic and physical environment for completing this project. Not only has this project helped me understand data and knowledge mining and machine learning better but also it has increased my awareness regarding how technology can be used to tackle global environmental challenges.

CHAPTER 1

INTRODUCTION

The paper project will be prepared and make the presentation at the end of the semester. The project must be based on the lecture of this course and it is individual assignment. By departmental policy, the discovery of plagiarism (i.e., copying from another student's project or other sources, including AI-generated writing) will result in a severe reduction of the relevant student's marks.

Font size 12 (Time New Roman). Line Spacing 1.5 line.

1.1 Project Background

The quick loss of forests represents a serious damage to the ecological balance, leads to biodiversity loss and undermines climate stability. Today, achieving sustainable planning to monitor deforestation, prediction on changes in forest cover and identifying the causal causes become essential for environmental balance. Since traditional methods of assessment generally do not have the ability to make proactive predictions, there is a pressing need for computational approaches based on data. In this project, I applies data mining techniques to discover relationships and patterns according to the data set. In this project, the Random forest classifier is mainly used as learning model for predicting deforestation and impacts of its. This project intends to meet the challenge of forecasting trends in deforestation and evaluating the related environmental impacts before critical thresholds are passed.

1.2 Objectives of the project

Objectives of making this project

- ❖ To identify important trends associated with deforestation by preprocessing and analyzing historical environmental and climatic variables.
- ❖ To investigate the performances of feature engineering of the prediction models.
- ❖ To use time-based data splitting to create reliable machine learning regression and classification models (such Random Forest and baseline models) while avoiding cases of data leaking.

- ❖ To assess model performance using feature appropriated rankings and comprehensive metrics in order to identify the main causes of deforestation.

1.3 Overview/Scope of the project

In this project, I mine a dataset with total records: 4,030 rows across 27 columns. In order to assess and forecast patterns mainly related to deforestation, this project focuses on using multi-year environmental data records from historical times to the present. In order to preserve linear integrity and avoid data leakage, the project scope includes thorough data preprocessing, managing missing values, and strict time-based data set division. I explored the relationships between deforestation metrics, socio-economic factors, climate change indicators, and biodiversity habitat loss using Python in Colab and data mining techniques. In order to guaranty strict academic standards, the project also includes the application of supervised machine learning methods, specifically Random Forest classifiers, as well as intricate classification tasks (predicting deforestation severity categories) and thorough evaluation analyzes.

1.3 Overview of the Methodology

In order to prepare the data for use, we first load the dataset and look for any abnormal data, strange formatting, or missing numbers. I carefully reviewed the features to drop anything that could cause data leakage like direct formula-derived scores and ensuring our models aren't cheating with hindsight information. Instead of doing a random shuffle(80-20)splitting, I split data using a time-based approach. The data set is specifically split into two parts: historical records up to 2015 are utilized as the training set to identify historical patterns, and data from years after 2015 is designated as the testing set to replicate actual future forecasts. This prevents the model from looking into the future and maintains the chronological sequence of events. To establish a performance benchmark, I began the modeling process with an interpretable baseline. In order to identify intricate patterns, I compared my data to more potent algorithms, particularly Random Forest Classifiers.

We did not employ accuracy measures for the regression jobs. So, I used the R-squared score to determine how well independent variables explained the variance in

deforestation, Mean Absolute Error (MAE) to see the average magnitude of our errors, and Root Mean Squared Error (RMSE) to heavily penalize larger prediction misses.

To ensure that our models were accurately identifying high-risk events rather than merely guessing the majority class, I followed the criteria for classification tasks, such as accuracy, precision, recall, and the F1-score.

1.5 Benefit of the project

Policymakers, conservation planners, and environmental researchers can all benefit from the project's conclusions. Stakeholders can create focused intervention plans to reduce ecological harm and advance sustainable land management by precisely identifying high-risk deforestation zones and key contributing elements (such as biodiversity indices and forest cover percentages).

CHAPTER 2

DATA PREPARATION AND PREPROCESSING

2.1 Dataset and Data Statistics

This dataset has 25 Numerical attributes and 2 Categorical attributes. The attribute Year is in Integer Format. The data covers a span of 31 years between 1990-2020. The dataset has a total of 4,030 records and 27 variables.

The Categorical Attributes are Entity and Region. The Numerical Attributes are: Year, DeforestationHa, AgriculturalLandPct, PopulationDensity, GDPPerCapita, RuralPopulationPct, PovertyRatePct, EmploymentAgriPct, ForestCoverPct, CO2EmissionsMt, LostCarbonSinkkt, BiodiversityImpactIndex, EnvironmentalImpactScore, PM25EmissionsTons, PM10EmissionsTons, PM25MeanExposureugm3, TemperatureAnomalyC, Precipitationmm, SPEIDroughtIndex, ExtremeHeatDaysCount, SpeciesHabitatLossPct, IUCNThreatenedSpeciesCount, BiodiversityIntactnessIndexPct, ProtectedAreaCoveragePct and ForestFragmentation_Index.

A preliminary data quality report have been carried out before pre-processing where missing data has been observed and handled and also structural integrity over all the 4,030 records has been made sure.

Table 2.1.1 Description of Dataset Attributes

Feature	Category	Definition / Data Type
Entity	Categorical (Object)	Country or geographic entity name
Year	Numeric(float64)	Year of observation from 1990 to 2020
Deforestation_Ha	Numeric(float64)	Deforestation area in hectares
Agricultural_Land_Pct	Numeric(float64)	Agricultural land percentage

Population_Density	Numeric(float64)	Population density metrics
GDP_Per_Capita	Numeric(float64)	Gross Domestic Product per capita
Rural_Population_Pct	Numeric(float64)	Rural population percentage
Poverty_Rate_Pct	Numeric(float64)	Poverty rate percentage
Employment_Agri_Pct	Numeric(float64)	Employment in agriculture percentage
Region	Categorical (Object)	Geographic region classification
Forest_Cover_Pct	Numeric(float64)	Forest cover percentage
CO2_Emissions_Mt	Numeric(float64)	Carbon dioxide emissions in megatons
Lost_Carbon_Sink_kt	Numeric(float64)	Lost carbon sink in kilotons
Biodiversity_Impact_Index	Numeric(float64)	Biodiversity impact index score
Environmental_Impact_Score	Numeric(float64)	Overall environmental impact score
PM25_Emissions_Tons	Numeric(float64)	PM25 emissions in tons
PM10_Emissions_Tons	Numeric(float64)	PM 10 emissions in tons
Temperature_Anomaly_C	Numeric(float64)	Temperature anomaly in degrees Celsius
Precipitation_mm	Numeric(float64)	Total precipitation in millimeters
SPEI_Drought_Index	Numeric(float64)	Standardized Precipitation-Evapotranspiration Index
Extreme_Heat_Days_Count	Numeric(float64)	Count of extreme heat days

Species_Habitat_Loss_Pct	Numeric(float64)	Percentage of species habitat loss
IUCN_Threatened_Species_Count	Numeric(float64)	Count of IUCN threatened species
Biodiversity_Intactness_Index_Pct	Numeric(float64)	Biodiversity intactness index percentage
Protected_Area_Coverage_Pct	Numeric(float64)	Protected area coverage percentage
Forest_Fragmentation_Index	Numeric(float64)	Forest fragmentation index score
PM25_Mean_Exposure_ug_m3	Numeric(float64)	PM25 mean exposure level in _ug_m3

Table 2.1.2(a) Statistics for Socio-Economic and Land Use Features

Feature Name	Count	Mean	Standard Deviation	Min	Median	Max
Year	4030	2005.00	8.95	1990.00	2005.00	2020.00
Deforestation_Ha	4030	92839.67	398185.30	0.00	4,940.00	6433770.00
Agricultural_Land_Pct	4030	39.70	22.18	0.49	40.84	85.46
Population_Density	4030	176.87	607.23	0.14	62.44	7918.95
GDP_Per_Capita	4030	12328.37	19137.47	98.80	4070.61	167187.16

Feature Name	Count	Mean	Standard Deviation	Min	Median (50%)	Max
Rural_Population_Pct	4,030	41.32	22.03	0.00	39.05	90.24
Poverty_Rate_Pct	4,030	26.28	12.23	0.00	23.10	69.40
Employment_Agri_Pct	4,030	25.32	21.43	0.11	18.58	84.53
Forest_Cover_Pct	4,030	30.37	22.06	0.00	28.64	96.23
CO2_Emissions_Mt	4,030	16.71	71.67	0.00	0.89	1,158.08
Lost_Carbon_Sink_kt	4,030	897.95	137.93	236.20	889.95	1,901.30
Biodiversity_Impact_ Index	4,030	0.54	0.46	3.00	0.54	0.94

Table 2.1.2(b): Descriptive Statistics for Climate, Emissions, and Biodiversity Features

Feature Name	Count	Mean	Standard Deviation	Min	Median (50%)	Max
PM25_Emissions_ Tons	4,030	25.33	8.91	14.00	24.00	120.00
PM10_Emissions_ Tons	4,030	3.69	9.01	1.00	1.00	48.91
PM25_Mean_Exposure _ug_m3	4,030	32.12	11.73	15.00	32.83	80.89
Temperature_ Anomaly_C	4,030	0.62	0.22	0.11	0.61	1.20
Precipitation_mm	4,030	897.95	137.93	236.20	889.95	1,901.30
Feature Name	Count	Mean	Standard	Min	Median (50%)	Max

			Deviation			
SPEI_Drought_Index	4,030	0.54	0.46	-3.00	-0.54	0.94
Extreme_Heat_Days_ Count	4,030	25.33	8.91	14.00	24.00	120.00
Species_Habitat_ Loss_Pct	4,030	3.69	9.01	1.00	1.00	48.91
IUCN_Threatened_ Species_Count	4,030	276.81	68.62	80.00	265.00	580.00
Biodiversity_ Intactness_Index_Pct	4,030	83.06	6.80	51.46	83.35	97.11
Protected_Area_ Coverage_Pct	4,030	12.06	4.92	2.88	11.71	41.63
Forest_Fragmentation _Index	4,030	31.73	18.36	5.00	31.48	98.00

Table 2.1.3: Dataset Data Quality

Data Quality Check	Result
Total Records	4030
Total Attributes	27
Missing Values	0
Duplicate Records	0

Table 2.1.3 demonstrates that the dataset contains no duplicates or missing data. Therefore, throughout the data preparation stage, neither missing value imputation nor duplicate deletion were required.

2.2 Data Preprocessing

To ensure data integrity, optimal model performance, and academic rigor for the deforestation dataset, a systematic preprocessing pipeline was implemented. This phase transforms raw data into a clean, structured format suitable for machine learning algorithms.

2.2.1 Handling Missing Values

During the initial data quality inspection, the 4,030 records in the deforestation dataset across 27 attributes were thoroughly examined to look for any missing or null entries. According to the study, all numerical and categorical features in the dataset have zero missing values. Hence, no missing value imputation technique was required. According to the observation, I didn't applied any method for handling missing values.

2.2.2 Data Transforming

The numerical features in the dataset are characterized by varying distributions, and some features are highly positively skewed. Specifically, Deforestation_Ha, CO2_Emissions_Mt, Lost_Carbon_Sink_kt, PM25_Emissions_Tons, PM10_Emissions_Tons features exhibit highly skewed distributions. Due to the presence of large variations between smaller and larger data in the above features, log1p transformation was used to minimize the influence of skewed distributions.

The dataset also includes categorical features like Region and Entity. As these features are categories, not numerical features, one-hot encoding was used to transform the categorical data into numerical form.

Thus, log1p transformation was an appropriate technique to use for the highly skewed numerical features, while one-hot encoding was an appropriate transformation for the categorical features since the machine learning models that were chosen require numerical features.

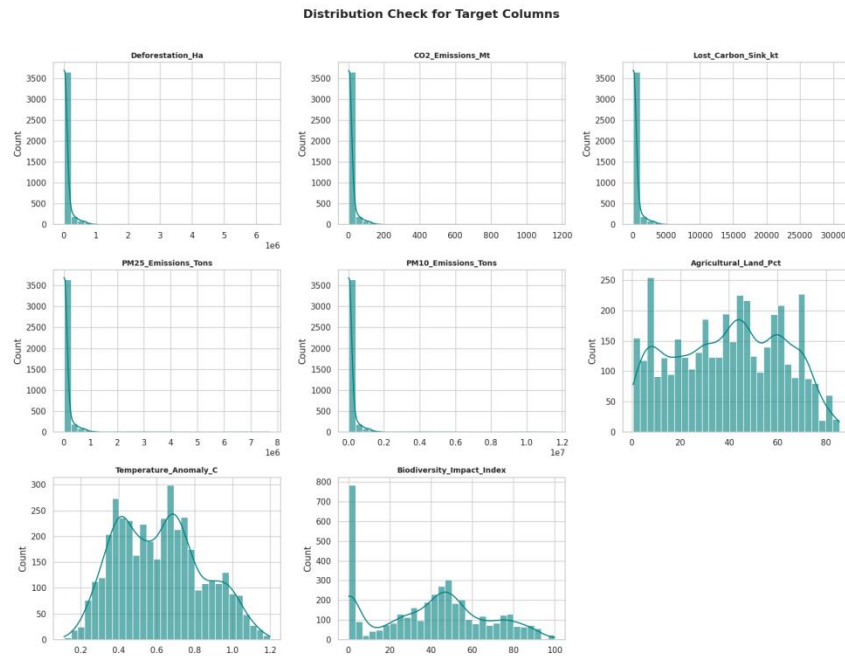


Figure 2.2.2.1 Feature Distributions and Skewness Analysis

2.2.3 Feature Selection

To determine the best variables that best explain the deforestation prediction using the information present within the features of this study, feature selection was utilized. The 19 variables from feature dependency information, associational-rule evidence, and correlation analysis comprised the candidate pool. Random Forest feature importance was used to rank each candidate feature from 1st to 19th.

Random Forest classification and regression evaluated several size of feature subset.

Tested features sets included 5, 8, 10, 12, and 15. Predictive performance evaluated using MAE, RMSE, R was used to select subset. The Top-13 feature model for Random Forest regression was the best of all others, giving MAE 16268.74, RMSE 44440.98, and R 0.9747, so it was used as the prediction model. With Neural Network, feature selection greatly improved prediction performance.

The Top-10 neural model gained R as 0.9227, and the original models gained as 0.8183 for regression.

However, the regression performance achieved by the Top-13 Random Forest was significantly better, and was finally chosen as the regression model. The Top-10

Random Forest model has improved the prediction for classification as well. Top-10 Random Forest classification model gains accuracy value 98.62, and Macro F1 value 98.61, whereas initial model 98.46, and 98.45.

Therefore, the Top-10 Random Forest was decided as the final classification model.

2.2.4 Data Normalization

To start, I'll cover data normalization, the purpose of which is vital to my dataset. Fundamentally, algorithms in machine learning, or data mining, can be biased towards features which may have wide ranging values (i.e comparing absolute land-area, or population-density to percentages, for instance, if these features were at all involved) and I don't want this! My intention is to scale all numerical feature-column values from an original range from zero up to another maximum (typically to between zero and one), while trying to preserve all ranges and therefore not discard any information.

I picked Min-Max Scaling (Normalization). Mathematical Explanation

$$\mathbf{X}_{scaled} = \frac{\mathbf{X} - \mathbf{X}_{min}}{\mathbf{X}_{max} - \mathbf{X}_{min}}$$

I Fit and transform the numeric data using the scaler to all variables are operating on the same range.

A small validation step at the end to demonstrate the results.

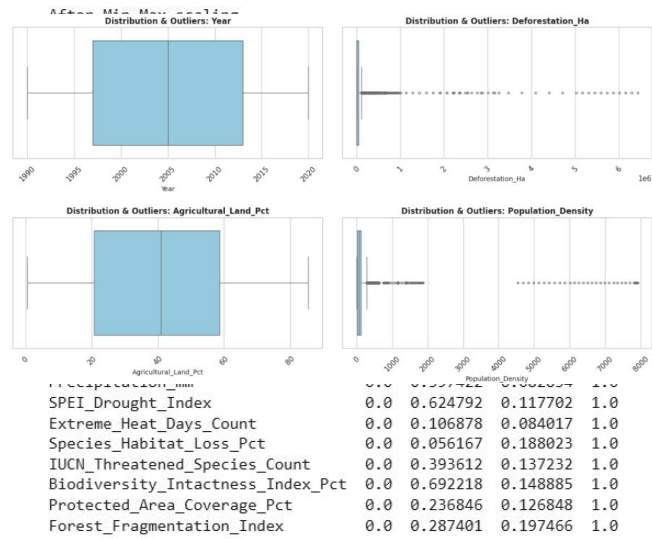


Figure 2.2.4.1 Feature after scaling (Min-Max scaling)

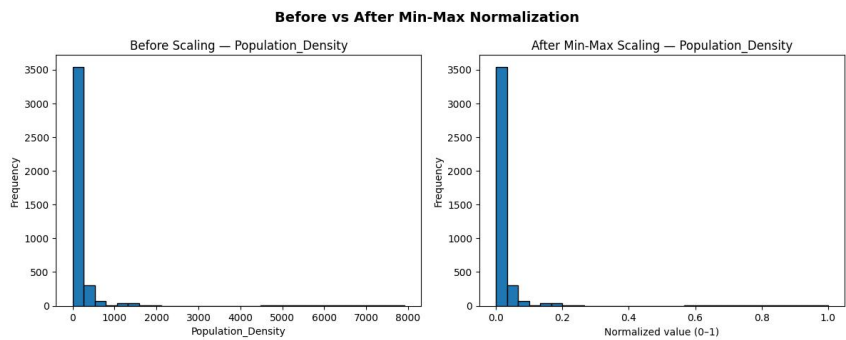


Figure 2.2.4.2 Comparison between the Population_Density feature before and after scaling (Min-Max scaling)

2.3 Data Visualization

It is an intermediary tool in between data preprocessing and predictive modeling which we are not concerned of the final model outcomes. Our interests here is to explore, search, identify trends and structure information embedded within the data such as distribution and relationships of variables and outliers, and map them into visual representation which would aid our feature selection.

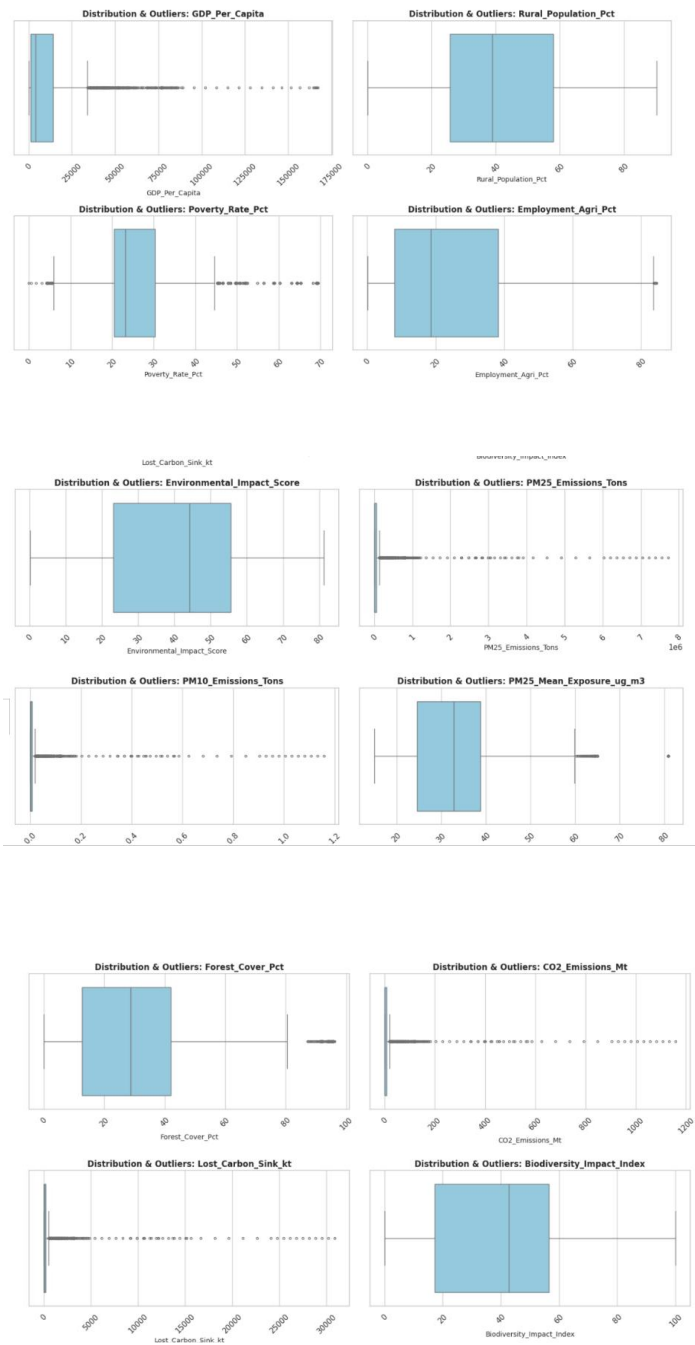


Figure 2.3. 1 A selection of numerical features shown on box plots in order to visualize the distributions and the detected outliers.

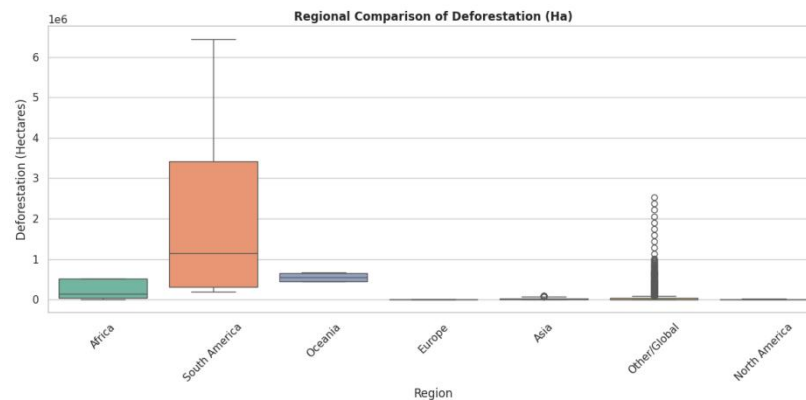


Figure 2.3.2 Box plot illustrating land deforestation (measured in hectares) for each region on the globe.

Figure 2.3.2 illustrates major differences in the volumes of deforestation. South America had both the greatest overall spread and largest overall range-signaling extremely high regional variation. Both Africa and Oceania display moderate spread, whilst regions such as Europe, Asia, and North America displayed extremely small range yet comparatively minute volumes.

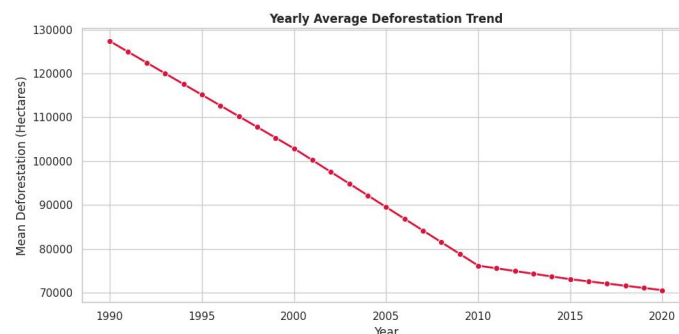


Figure 2.3.3 Line graph showing the global annual average trend in deforestation during the 1990 to 2020 period.

Figure 2.3.3 demonstrates that throughout this thirty-year period there was a sustained, long-term decrease in the average rate of global deforestation. The gradient of the trend line show quite a rapid, sustained decline throughout the 1990 to 2010 period, prior to leveling out at a lower rate of decline through to 2020.

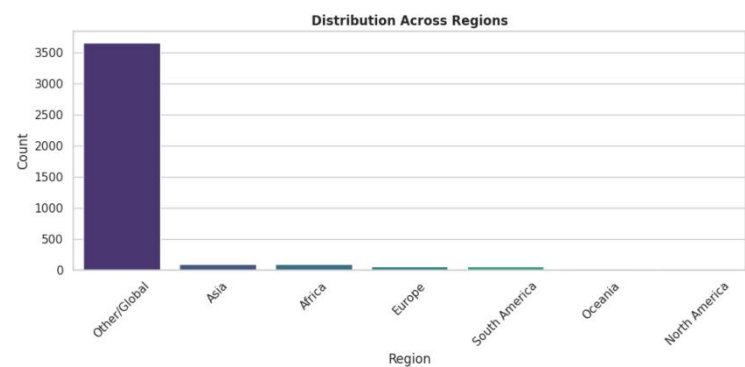


Figure 2.3.4 A bar chart illustrating the frequency distribution of data throughout geographical locations.

Figure 2.3.4 bar chart shows that the records are not distributed equitably throughout. "Other/Global" accounted for more than 3500 in number while others such as "Africa", "Asia", "Europe", "Oceania", "North America" and "South America" account for almost zero.

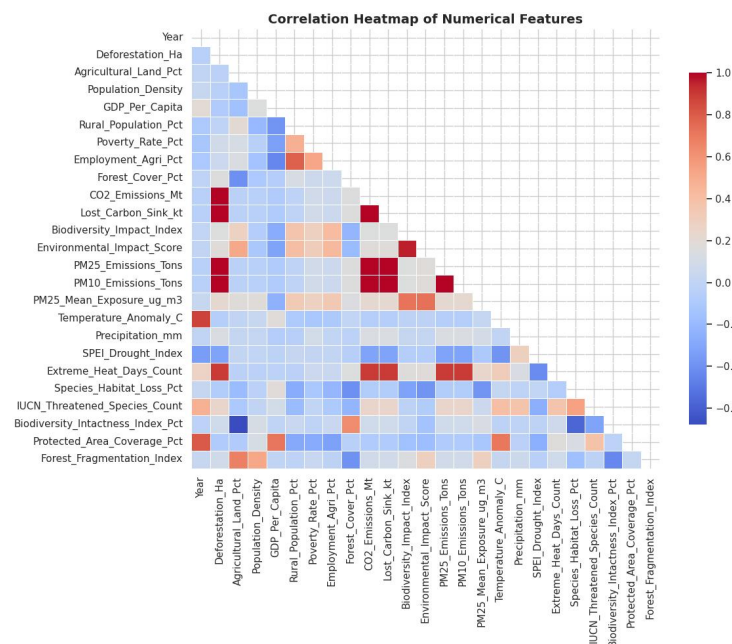


Figure 2.3.5 Ccorrelation heat-map

Shows are all of numerical variables correlated with each other, and plotted on color scale ranging from blue which mean negative correlate to red meaning positive. The chart presents Collinearity and correlations in the features space - For example,

correlations between several different emissions and loss of carbon sink, these are significantly highly positive thus helps in selection of feature.

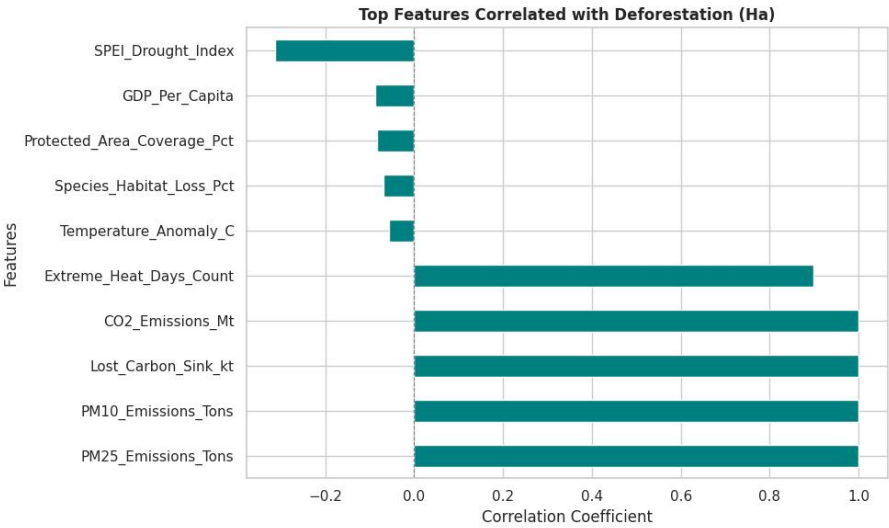


Figure 2.3.6 Top features correlated with Deforestation(Ha)

CHAPTER 3

APPLIED METHODOLOGIES AND MINING

3.1 Descriptive Mining

Descriptive mining can be defined as exploring data in order to find interesting groups, natural groupings and "hidden" relationships but without looking for a target class. It's about putting together exploratory results.

3.1.1 Association Rules Mining

3.1.1.1 Frequent Itemsets

The most occurring paired condition includes low poverty and low habitat loss, together this occurred in ~58.4% of all data records. Combined, low extreme heat day and low habitat loss occurred in 39% of the data set. Low threatens species count and low habitat loss occurred in 33.7% of the data records. The remaining columns outline standard conditions that are seen consistently about 33% to 34% of the time: average yearly conditions and the baseline levels for a wide variety of other conditional information including a lower drought index and a lower than average change in temperature.

1. FREQUENT ITEMSETS TABLE (Top 15 by Support)

	support	itemsets
0	0.922189	(Species_Habitat_Loss_Pct_Low)
1	0.662426	(Poverty_Rate_Pct_Low)
2	0.584615	(Poverty_Rate_Pct_Low, Species_Habitat_Loss_Pc...
3	0.433136	(Extreme_Heat_Days_Count_Low)
4	0.392899	(Extreme_Heat_Days_Count_Low, Species_Habitat_...
5	0.346154	(Year_Medium)
6	0.346154	(Year_Low)
7	0.336686	(IUCN_Threatened_Species_Count_Low)
8	0.336686	(IUCN_Threatened_Species_Count_Low, Species_Ha...
9	0.336095	(Temperature_Anomaly_C_Low)
10	0.335799	(SPEI_Drought_Index_Low)
11	0.335799	(Population_Density_Medium)
12	0.334911	(SPEI_Drought_Index_Medium)
13	0.334911	(Temperature_Anomaly_C_Medium)
14	0.334320	(Precipitation_mm_Low)

Figure 3.1.1.1 Top 15 frequent item-sets based on support

3.1.1.2 Support vs Confidence

I draw plot to see together three measures of our association rules to easily analyze the quality and strength of our rules using a scatter plot. The X-axis represent values for Support, the Y-axis represent Confidence (how reliable the rule is), and the color strip to the right represent Lift (how much higher the probability of the consequent given the antecedent is relative to the independent probabilities). Majority of our rules are found grouped at lower values of support with a value of about 0.1 in the left-hand cluster (meaning a pattern occurs 10% of the time) yet have an accuracy that reaches all the way to 1.0 or 100%. We observe a cluster of high lift scores with the greener and yellow dots.

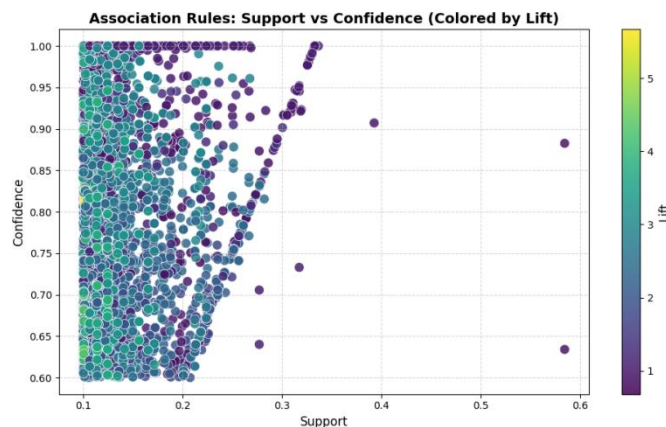


Figure 3.1.1.2 Visual representation of the extracted associations: Support vs Confidence with different colors representing 'Lift' strength for these rules.

3.1.1.3 Association Rule Mining

Aim to retrieve strong relationships hidden within the predictors and not caused by unintended leakage from the continuous target variable (Deforestation_Ha), association rules were tailored specifically for discrete predictor attributes rather than continuous target bins. Discretization Procedure: All continuous predictor variables were binned into three equal count, ordinal bins (Low, Medium, High) using quantile-based binning (pd.qcut). To avoid data leakage: the continuous target variable (Deforestation\Ha) was explicitly excluded from the basket of the association rules (to prevent thousands of items of value 0 deforestation and endless feedback loops).

Support and Confidence parameters: Frequent itemsets were found using min support=0.10(10%), and then strong association rules found were narrowed by min confidence of=0.60(60%).

Key variables assessed: support, confidence, Lift, leverage and conviction; high- Lift associations highlighted.

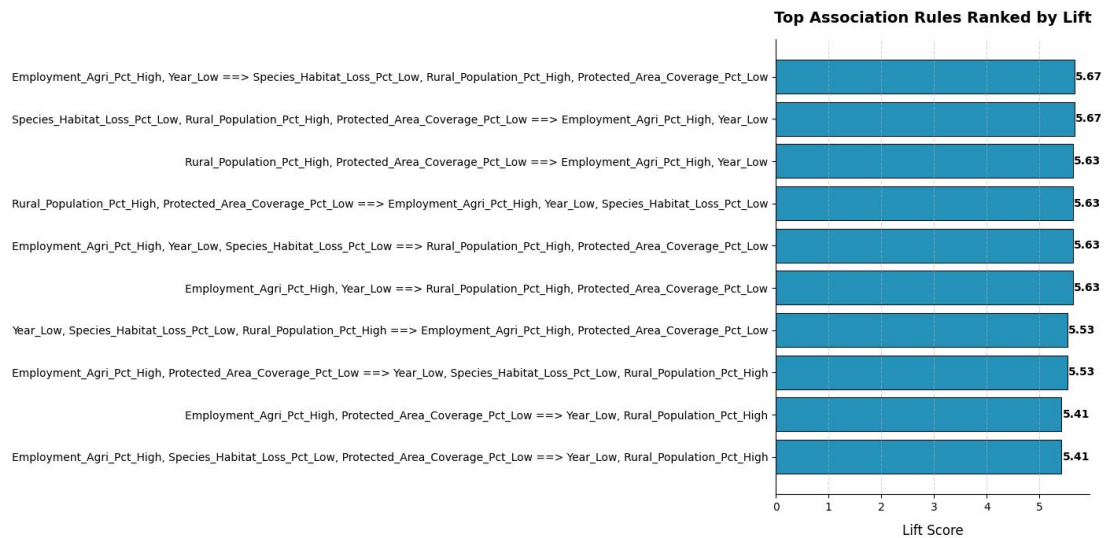


Figure 3.1.1.3 Top 10 association rule sorted by the Lift score (≥ 5.4)

Interpretation of Top Lift Rules

The top 10 rules derived had very high lift values ranging from 5.41 to 5.67, suggesting a very strong positive association between the antecedent and consequent terms within each rule set.

3.1.2 Clustering Analysis

Below is ten features which were chosen from the dataset for the clustering process:

“Deforestation_Ha,AgriculturalLandPct,Population_Density,GDPPerCapita,ForestCoverPct,CO2EmissionsM,tBiodiversityImpactIndex,TemperatureAnomalyC,Precipitation_mm,SpeciesHabitatLoss_Pct”.

3.1.2.1 K-Means

K-Means is an algorithm in cluster analysis that partitions the data into a pre-specified number of groups, K. We have trained the model on our scaled dataset. The KMeans module was utilized with a state of randomstate= 42 so that each run of the algorithm produced the same result and n-init=10 so that it has performed all-possible initial cluster centroids, so that it can yield better result without getting trapped in bad local minimum and has been evaluated over each iteration to have the least score, and then has been run for each it

3.1.2.2 Selection of K using Silhouette Score

Determining an optimal K to best divide the regions or countries by varying values between 2 and 10 and evaluating Inertia and Silhouette scores produced by each value of K.

Table 3.1.2.2 Evaluation Results for Candidate K Values

	K	Inertia_WCSS	Silhouette_score
0	2	1293.5396	0.2520
1	3	1066.2062	0.2247
2	4	898.7392	0.2317
3	5	786.7744	0.2495
4	6	714.7177	0.2437
5	7	665.5290	0.2303
6	8	625.8833	0.2435
7	9	594.0440	0.2361
8	10	564.6515	0.2363

After performing several experimentations, $K = 2$ has been determined to be the optimum value which gives the maximum Silhouette score of 0.2520, the highest value in range, thereby indicating that the countries are divided efficiently into 2 categories.

3.1.2.3 Cluster Profiles

Below is the list showing the characteristics of the two chosen clusters after the K value is determined.

Cluster Distribution

Table 3.1.2.3.2 Cluster Distribution

	Count	Percentage
Cluster		
0	1992	49.43
1	2038	50.57

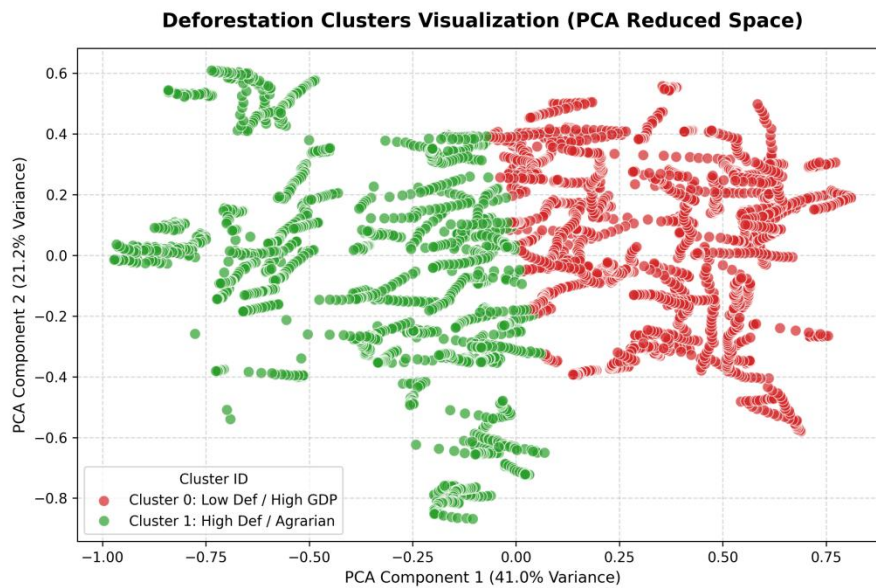


Figure 3.1.2.3.2 PCA cluster scatter

3.1.2.4 Cluster Rules and Interpretation

The following IF-THEN Decision Rules can be constructed based on the clustered attributes in relation to the global average data points to obtain information in interpretable way:

Cluster 0 Decision Rule:

IF (DeforestationHa = LOW) AND (AgriculturalLandPct = LOW) AND (PopulationDensity = HIGH) AND (GDPPerCapita = HIGH) ... THEN Cluster = 0

Cluster 1 Decision Rule:

IF (DeforestationHa = HIGH) AND (AgriculturalLandPct = HIGH) AND (PopulationDensity = LOW) AND (GDPPerCapita = LOW) ... THEN Cluster = 1

Interpretation of rules

The areas within this cluster (Cluster 0) belonged to developing countries with very high GDP per capita, a highly urbanized area (population density) and with a very low amount of land dedicated to agriculture as well as not deforested with characteristics that show an environment friendly region. The countries under this

cluster (Cluster 1) were poor and agrarian countries with a low GDP and with less population density but with a large amount of land for agriculture, greatly deforested land and high emission of CO₂ and carbon gases (a low value for the impact diversity).

3.1.2.5 Cluster Evaluation

Table 3.1.2.5 Cluster Evaluation

Metric	Value
Overall Silhouette Score	0.251964702
Cluster 0 Silhouette	0.166964506
Cluster 1 Silhouette	0.335046346

Silhouette analysis showed overall index 0.252 which showed weak-moderate structure of clustering and overlap was found. Cluster 0 showed lower and looser cohesion and separation 0.167 while Cluster 1 had better defined clusters showing better internal cohesiveness with 0.335.

3.2 Modeling

For accurate forecasting and categorization of deforestation, predictive modeling is employed. This section delineates the algorithms applied, their configurations, and the metrics used for evaluation in assessing deforestation levels Deforestation\Ha and classifying regions by risk status.

3.2.1 Applied Algorithms

3.2.1.1 Random Forest Regression

Random Forest Regression builds an ensemble of multiple decision trees during the training phase. The final output for a given input is the average of the individual tree predictions. This approach often reduces overfitting and can handle complex, non-linear relationships within the data. The Random Forest Regression builds a collection

of decision trees during training. And the prediction for a specific input is the average value of all predicted outputs by decision trees. This can help avoid over-fitting, especially if data have complex, non-linear relationship.

3.2.1.2 Neural Network Regression

Neural Network Regression models, specifically a multi-layer perceptron, are used to capture intricate non-linear patterns present among climate anomalies, socio-economic indicators, and habitat degradation predictors influencing deforestation.

3.2.1.3 Random Forest Classification

To label risk categories-converting the continuous deforestation target into a binary variable, where the classification threshold is based on the median of DeforestationHa from training dataset (DeforestationHa median corresponds to class 0 \[Low] and Deforestation_Ha>median corresponds to class 1 \[High]). The models are evaluated using Accuracy, Balanced Accuracy, Macro Precision, Macro Recall, and Macro F1-score. Confusion matrix analysis is also performed to thoroughly examine classification results.

3.2.2 Model Setup

All predictive models adhere to a strict temporal train/test split (e.g., data up to Year > 2015 for training and Year ≤ 2015 for testing) to circumvent data leakage and maintain reliable out-of-sample generalization. Similarly, target downstream effects or metrics directly derived from them (e.g., CO2 emissions, loss of carbon sinks) are not used as predictors in order to achieve a practical assessment.

Fields CO2EmissionsMt, LostCarbonSinkkt, PM25EmissionsTons, PM10EmissionsTons, EnvironmentalImpact_Score are deliberately left out of the prediction pipeline to prevent target leakage and look-ahead bias.

Classification threshold is based on the median of DeforestationHa from training dataset (DeforestationHa median corresponds to class 0 \[Low] and Deforestation_Ha>median corresponds to class 1 \[High]).

Class distribution

Table 3.2.2.1 Class distribution of the binary deforestation risk target across training (Year $\leq$ 2015\$) and testing (Year $>$ 2015\$)

	Train count	Test Count
Low	1690	353
High	1960	297

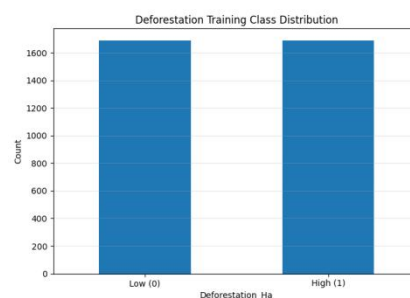


Figure 3.2.2 LOW and HIGH Class Distribution

Neural network will be built and trained using Tensorflow and Keras. features are then scaled using Min-Max scaling, fitted solely on the training data to avoid future information contamination. Missing values are filled with the training data median.

3.2.3 Baseline Model

3.2.3.1 Deforestation Regression Baseline Model (Random Foreset)

This is a Baseline Model that trained on all the features without leakage and predicts DeforestationHa, a highly right-skewed and zero-concentrated variable. During training, $\log(1+X)$ is used to transform DeforestationHa, and during prediction the outcome will be anti-transformed . A Random Forest Regressor on the full set of predictors (i.e. No leakage). The configuration of model are `ninamitors = 300`, `maxdepth = 15` and minimum samples at each leaf: `Minsampleleaf = 2`, shown in box.

Target variable right skewed, has been log transformed and so on. MAE, RMSE, R2 on the test (Year>2015) data were assessed model performance.

Baseline of Random Forest Classification on all variables that are not leaking with balanced class weights on to analyze two classification risk level for deforestation. Values, over the course, in natural binary classification, are segmented on a threshold based solely on training set median in order to prevent leakage of data. In control of class imbalance and of stable result on low and high values respectively, comprehensive analysis is conducted. Using Accuracy, Balanced Accuracy, Macro Precision, Macro Recall, Macro F1 score, evaluation will be conducted on models' predictive strengths.

3.2.3.2 Neural Network Baseline Model

Reframed problem into potentially classification problem from continuous regression by viewing the problem from different angle. Classification Definition, Classes Since it wasn't logical to define classification by arbitrary threshold value for segmentation purposes, I used 'realistic' classification defined only by training data (trainmediandefo), fixed benchmark, not close to leakage of any sort in data for training. All rows, if $\leq$ median class would get labeled as 'Low Deforestation'; $>$ median class as 'High Deforestation'. Built Neural Network I implemented a relatively simple Sequence neural network classification model: input layer sized by the number of features, two densely connected neural network hidden layers with ReLu activation and dropout layer between layers.

To avoid overfitting of model (0.2, 0.1 dropout between layers then sigmoidal activation to give predicted class probability for true binary class).

Built neural network & Compiled Adam with Binary cross-entropy, trained the neural network by batch_size of 32 for 150 epoch. Split to 85% data for training and 15% data for testing with dynamically used callbacks to speed up network training. Evaluated all prediction results against computed labels from the defined neural network and reported accuracy, precision, recall and f1 score in the performance table while distinguishing identified "High Deforestation" instances against prediction outcomes.

3.2.4 Proposed Model

The developed model comprises of a sequential implementation to avoid the leakage and use upmost predictive efficiency in the developed approach. The model used quantile based binning to segment the continuous predictor variables and further only the features attributes were used within the Apriori model to find hidden associations between various predictor variables. The continuous target variable was always segregated during the process and candidates features were synthesized out of the Spearman correlations, Mutual Information scores and the association rule. Selected subset of features were examined against the tree based permutation importance and time based validation set to ensure the out of sample performance.

CHAPTER 4

EVALUATION

4.1 Performance Metrics

Mean Absolute Error (MAE) provide insight on the "average" error in predictions in hectares. **Coefficiency of Determination (R²)**. Represents fraction of variability in dependent variables can be expressed in terms of dependent variables selected.

Metrics Classification (target is a category/ categorical prediction task) is **Accuracy & F1-Score**. Accounts for overall correctness and balance for all values. This metrics should prove very helpful to both Binning the continuous variable as target, or categorical variables created under risk evaluation.

ROC-AUC : Models discriminatory power between risk levels; The Receiver Operating Characteristic AUC curve assesses the performance or the discriminative ability between groups/ risks assigned.

Association Rule Metrics: Support & Confidence. Measure based on finding relevant item sets and asses whether rules discovered indicate genuine structural interrelationships or by pure chance.

Lift : takes precedence when value of it for an association rule is detected to be significant in order for the association rule (antecedent AND consequent is occurring) not to be detected by pure coincidence.

It's symmetrically treating both classes. Macro-averaging measures and evaluate for all the categories(The two binary categories "Low" and "High" deforestation types separately, and take unweighted average). So, any of minor class or class of interest has an equal opportunity being examined, no weighted by the accuracy at the major class only.

It's dealing with class imbalance issue. Many environmental data has a class distribution biased with some of thresholds. Just evaluate using general accuracy (same as micro-averaging) can be trickily high although the model just learn the commonest class correctly, and micro-average can be a bad signal. While macro-metrics' evaluation clearly shows that performance is not good in one of them, that shows any category of prediction performs bad.

It's analogous to the calculation of Balanced Accuracy (which calculating the average of recall of all categories), so it is a suitable metric for statistical partnership with Macro-precision, macro-recall and macro-F1 score, the indicator that evaluate correctly how well we are handling risk among several classes without bias.

4.1.1 Comparison of Random Forest Regression Baseline and Feature Optimized Model Performance

The evaluation section outlines a comparison and performance data of the Random Forest Deforestation Baseline model and the Random Forest Top 13 Features optimized model. To evaluate the performance MAE, RMSE and R^2 metrics are used. The lower MAE and RMSE metrics suggest greater accuracy. Higher R^2 suggests the greater proportion of variability.

Table 4.1.1 Top 13 selected features

1. PM25_Mean_Exposure_ug_m3
2. Biodiversity_Impact_Index

3. Forest_Cover_Pct
4. Extreme_Heat_Days_Count
5. Environmental_Impact_Score
6. Population_Density
7. Agricultural_Land_Pct
8. Poverty_Rate_Pct
9. Employment_Agri_Pct
10. Forest_Fragmentation_Index
11. IUCN_Threatened_Species_Count
12. GDP_Per_Capita
13. Temperature_Anomaly_C

Table 4.1.1.2 Comparison of Random Forest Regression Baseline and Feature Optimized Model Performance

		Model	MAE	RMSE	R2
0	Random Forest Deforestation Baseline		17268.270378	49053.578255	0.969184
1	Random Forest — Deforestation Top 13 Features		16268.744103	44440.982416	0.974707

The optimized feature space model was able to decrease MAE by approximately 1000 units and improve the RMSE over the basic model, signifying that it is better at penalizing big discrepancies in error. Also, the optimized features still hold about 97.47% of the variability in the target variable.

4.1.2 Comparison of Random Forest Classification Baseline and Feature Optimized Model Performace

The evaluation of classification is done comparing the performance of Random forest baseline model with the tuned model on top 10 features.

Table 4.1.2 Comparison of Random Forest Classification Baseline and Feature Optimized Model

	Model	Accuracy	Balanced_ Accuracy	Macro_Pre cision	Macro_Re call	Macro_F1
0	Baseline	0.984615	0.984767	0.984269	0.984767	0.984508
1	Classificati on Top 10 Features	0.986154	0.986184	0.985929	0.986184	0.986054

The evaluation on the classification performed to compare the baseline RF classifier and the improved one with 10 features chosen by highest scores. The baseline classifier presented accuracy of 0.9846 with balanced accuracy of 0.9848 and macro F1 score of 0.9845. The improved classifier (after feature selection) shows minor

gains in all classification metrics and reached an accuracy of 0.9862 with balanced accuracy of 0.9862 and macro F1 score of 0.9861.

4.1.3 Comparison of Random Forest Regression Baseline and Feature Optimized Model

Table 4.1.3 Comparison of Random Forest Regression Baseline and Feature Optimized Model

	Model	MAE	RMSE	R2
0	Random Forest Deforestation Baseline	17268.270378	49053.578255	0.969184
1	Selected RF — Top 10 Features	17354.253160	49309.292511	0.968862

This first method of feature-engineering using just Association rule derived feature resulted into degraded predictive model compared to baseline model. Hence, reconsider the feature selection approach. Used Association Rule for information and identified important feature from Spearman Correlation, Mutual Information. Sorted the candidates using Random forest permutation feature importance and the selection of number of feature selection by validation along time-series.

4.1.4 Comparison of Random Forest Classification Baseline and Feature Optimized Model

Table 4.1.4 Comparison of Random Forest Classification Baseline and Feature Optimized Model

	Model	Accuracy	Balanced_Accuracy	Macro_Precision	Macro_Recall	Macro_F1
0	Baseline	0.984615	0.984767	0.984269	0.984767	0.984508
1	Classification Selected Top 10 Features	0.984615	0.984767	0.984269	0.984767	0.984508

With comparing Baseline RF and Selected RF (using all available valid features) it's evident that the classification performance was exactly the same in all evaluation metrics, where Macro F1 of 0.9845 and total accuracy of 98.46% were observed.

While initial AR mining gave helpful explorations baskets, trying to use AR in a strict way to filter predictors in regression for a continuous variable such as Deforestation_Ha has failed as the values were too granular due to large number of discrete groups in variable; A new hybrid selection method combining Spearman's correlation, Mutual Information and Random Forest permutation importance allowed us to reach good quality variable selection in the initial set of predictors.

The key takeaway here is not that a new set of features produces more accuracy, but it builds a simpler model with same best possible metrics. It means that the data in selected features were redundant and contained the same type of information, which our hybrid feature selection successfully extracts.

4.1.5 Comparison of Neural Network Regression Baseline and Feature Optimized Model

Table 4.1.5 Comparison of Neural Network Regression Baseline and Feature Optimized Model

	Model	MAE	RMSE	R2
0	Neural Network — Deforestation Baseline	25051.795327	119119.308499	0.818284
1	Optimized model	19232.177025	77704.839570	0.922674

4.1.6 Comparison of Neural Network Classification Baseline and Feature Optimized Model

Table 4.1.6 Comparison of Neural Network Classification Baseline and Feature Optimized Model

Model	Accuracy	Macro_Precision	Macro_Recall	Macro_F1
Baseline	0.9677	0.9650	0.9700	0.9674
Optimized Model	0.9754	0.9750	0.9755	0.9752

4.2.2 ROC Curve and AUC

The ROC curve was a method used to classify ability to separate low and high deforestation class over various decision thresholds and each classification model is the corresponding value for the AUC. They measure the overall class separability for each classifier. Larger AUC means better separating the low and high deforestation class, so higher AUC mean of model perform classification discrimination on test period more effectively For the deforestation classification model, ROC-AUC was

applied to compare the effectiveness of models, in which RF Top 10 scored best AUC of 0.9988; Association Hybrid, RF Baseline, and Neural Network have accuracy nearly as high as that of RF Top 10 (0.9986, 0.9986, 0.9980 separately)

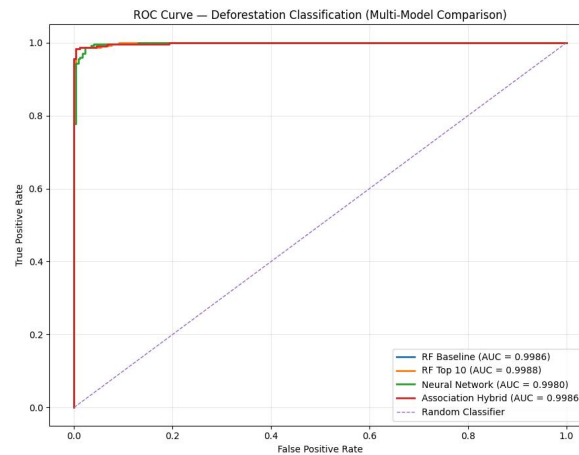


Figure 4.2.2 ROC Curve and AUC

4.2.3 Cross Validation

The hold-out data set (Year > 2015) is not included in cross validation. 5-fold K-Fold CV on the training data is used for regression. Target is still $\log_{10}(\text{Deforestation_Ha})$ as models are trained like that with Random Forests for regression. CV scores reported below report R, MAE and RMSE on the target (log-scale) and can not be used in comparison to the final test scores on the original scale with R, MAE and RMSE. Among the models in the comparison of cross-validation performances: "RF Classification - Top 10" perform the best and holds the highest accuracy(0.9923), balance accuracy(0.9923), macro F1(0.9923) and AUC(0.9998). "Hybrid Association + RF Classifier" and "RF Classification - Baseline" obtained also accuracy > 0.99. The std values are very low (lower or equal to 0.0032). This suggest the stability across the different folds for these models and can help reducing risk of over-fitting.

Table 4.2.3.1 Regression Cross Validation

Model	MAE (mean $\pm$ std)	RMSE (mean $\pm$ std)	R^2 (mean $\pm$ std)
Association Hybrid Regression Model	0.0668 $\pm$ 0.0068	0.1908 $\pm$ 0.0555	0.9979 $\pm$ 0.0012
RF Regression — Top 10	0.0668 $\pm$ 0.0068	0.1908 $\pm$ 0.0555	0.9979 $\pm$ 0.0012
RF Regression — Baseline	0.0745 $\pm$ 0.0071	0.1940 $\pm$ 0.0595	0.9978 $\pm$ 0.0013
Neural Network Regression — Baseline	0.1132 $\pm$ 0.0085	0.1516 $\pm$ 0.0122	0.9987 $\pm$ 0.0002

RF Regression - Baseline: Top performing / basemodel. Takes advantage of Random Forest algorithm using the full, untuned set of features.

RF Regression - Top 10: Feature-engineered / Optimized model. Explores feature subsetting by taking only the 10 best, chosen, features.

Association Hybrid Regression Model: hybrid / Optimized model. Includes features/outputs from Association Rule Mining or influences the feature set by these outcomes.

Neural Network Regression - Baseline: Top performing / basemodel. Uses the simplest MLPRegressor on a scaled full feature-set without parameter optimization and without considering model's structure on whole set of features.

Table 4.2.3.2 Classification Cross Validation

Model	Accuracy (mean $\pm$ std)	Balanced Acc (mean $\pm$ std)	Macro F1 (mean $\pm$ std)	AUC (mean $\pm$ std)
RF Classification — Top 10	0.9923 $\pm$ 0.0032	0.9923 $\pm$ 0.0032	0.9923 $\pm$ 0.0032	0.9998 $\pm$ 0.0001
RF Classification — Baseline	0.9914 $\pm$ 0.0032	0.9914 $\pm$ 0.0032	0.9914 $\pm$ 0.0032	0.9998 $\pm$ 0.0001

Hybrid Association + RF Classifier	0.9905 ± 0.0031	0.9905 ± 0.0031	0.9905 ± 0.0031	0.9997 ± 0.0001
Neural Network (MLP) — Baseline	0.9867 ± 0.0081	0.9867 ± 0.0081	0.9867 ± 0.0081	0.9990 ± 0.0009

Classification Models Summary

RF Classification - Baseline: Baseline model. A basic Random Forest classifier on the entire, unreduced feature set.

RF Classification - Top 10: Feature-engineered / Optimized model. A Random Forest classifier limited to only the top 10 picked features.

Hybrid Association + RF Classifier: Hybrid / Optimized model. A combination of the Association rule-mining process with the RF classifier.

Neural Network (MLP) - Baseline: Baseline model. A basic Multi-Layer Perceptron classifier (MLPClassifier) with default baseline settings on the entire feature set.

4.2.4 Findings

The models chosen were based on the test set performance as well as cross validation performance. With respect to the performance metrics, it can be noted that the following results were obtained:

For regression problems, the top 13 Random Forest features provided the best performance, which was achieved at a MAE value of 16,268.74, RMSE value of 44,440.98, and R-value of 0.9747.

For classification problems, the top 10 Random Forest features provided an outstanding performance, with an accuracy of 98.62%, a macro F1 score of 98.61% and an AUC value of 0.9988.

Model Stability - The cross validation stability with regard to all the folds was great, since the standard deviations were very small (less than 0.0032).

CHAPTER 5

CONCLUSION

In this study, trends in global deforestation as well as environmental impacts connected to the problem were analyzed and predicted using data mining and machine learning. Applying comprehensive data preprocessing, time-based data partition, association rules mining, clustering, and prediction methods, including Random Forests and neural networks on a multivariate dataset for the years 1990-2020 with 27 variables, the study succeeded in building highly effective models, which showed an R-squared of 0.9747 for the top-13 Random Forest regression model and accuracy of 98.62% for the top-10 Random Forest classification model.

5.1 Advantages of Using This Project

Proactive Environmental Management: By revealing deforestation risk zones as well as the main factors that cause the destruction of forests, the results of this study may be applied to planning and implementing necessary measures for mitigation of adverse effects.

Prevention of Target Leakage: The lack of look-ahead bias and target leakage was achieved by applying the time-based train/test split (using only data prior to 2015 for training and the data after 2015 for testing) and avoiding using features derived directly from target, such as CO2 emissions and carbon sink loss.

High Prediction Accuracy: The combination of Spearman correlation, mutual information, and permutation importance in Random Forest allowed the creation of models with high predictive accuracy and minimal redundancy in the feature space.

Comprehensive Descriptive and Predictive Mining: The analysis of both descriptive and supervised aspects (association rules mining and K-Means clustering combined with regression and classification) revealed the socio-economic, climate-related, and biodiversity drivers of deforestation.

5.2 Disadvantages of using this project/ Limitation

Drawbacks and Restrictions Associated with the Project

Geographical Data Imbalance: There is significant skewness in terms of geographical distribution of the records, whereby most records are classified as “Other/Global,” while fewer records belong to individual regions. This imbalance limits the level of granularity that is needed to produce localized predictions.

Granularity Problems in Association Rule Mining: The use of association rules on continuous data involved the use of quartiles for the discretization of the variables. Continuous target variable such as Deforestation_Ha could not be filtered using the association rules because of high granularity levels.

Historical Basis of Predictive Models: The models used in the project are based on historical data until year 2015. Therefore, any unforeseen ecological disaster or policy changes beyond the historical trends will impact the performance of the model in predicting future events.

REFERENCES

REFERENCES

[1] Food and Agriculture Organization of the United Nations (FAO), *Global Forest Resources Assessment: Main Report*, Rome, Italy: FAO, 2020. [Online]. Available: <https://www.fao.org/forest-resources-assessment/>

[2] The World Bank, *World Development Indicators: Agricultural Land, Population Density, and GDP per Capita*, Washington, DC: World Bank. [Online]. Available: <https://datatargets.worldbank.org/>

[3] International Union for Conservation of Nature (IUCN), *The IUCN Red List of Threatened Species*, Gland, Switzerland: IUCN. [Online]. Available: <https://www.iucnredlist.org/>

[4] World Resources Institute and Global Forest Watch, *Interactive World Forest Loss and Deforestation Monitoring Data*, Washington, DC: WRI. [Online]. Available: <https://www.globalforestwatch.org/>

[5] The-Morgan-Kaufmann-Series-in-Data-Management-Systems-Jiawei-Han-Micheline-Kamber-Jian-Pei-Data-Mining.-Concepts-and-Techniques-3rd-Edition-Morgan-Kaufmann-2011

[6] <https://www.coursera.org/articles/random-forest>

